Project Milestone P2

Analysis and High-Level Design

BookBridge

Team	Team [6], section [.003]
Document version	1.0 (baseline for P2; grows from your P1 version 0.x)
Date	8th Nov 2026
Repository URL	https://github.com/rajitplay/CS-3354-Team-6
Git tag for this milestone	p2 (points at the commit that contains this exact PDF)
Submitted on Canvas by	Derrick Greene

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0 Revision History, Contribution Record and Changes Since P1

All three tables below are graded. The contribution record carries the individual 20% of this milestone and must agree with your repository history.

Version	Date	Author	Summary of change
0.3	10/18/2026	Steven Law	Updated Sections 0.1, 3.2–3.5, 4.3–4.5, and AI use log
1.0	11/08/2026	Team 6	P2 baseline: analysis and high-level design added

0.1 Individual contribution record (P2)

Team member	Sections and artifacts this person produced for P2	Evidence in the repository (commit hashes, PR numbers)	Reviews given (PR numbers)
Rajit Ravindran	Section 6.1-6.4, 6.6	11/8/2026 1:20 AM – 11/8/2026 6:34 PM	
Derrick Greene	Completed the Stripe sandbox payment feasibility test for BookBridge (R2), including a successful $25 test transaction. Documented the results in docs/payment-feasibility.md. Coordinated team communication, followed up with members about deadlines and responsibilities, and updated the README and team plan based on P1 feedback. I also completed Sections 10 and 11 of the P2 document, including the project plan, risks, and open questions. I updated my contribution record, corrected several inconsistencies in the document, completed the revision history, and removed the template	docs/payment-feasibility.md — commit af97f4c README.md — updated project information and team roles docs/team-plan.md — updated working agreement and team responsibilities Microsoft Teams — communication and follow-up messages with team members. README.md — commits f08a490, 30e99a0, docs/team-plan.md — commit d06bd84	

	instruction boxes before submission.		
Abran Garicia	Completed ADR-01, ADR-02, and ADR-3 in section 7.1		
Steven Law	Sections 3.4 and 3.5 (updated use case list and use case diagram with UC-07), 4.1 to 4.3 (use cases selected, expanded UC-03, UC-04 and UC-01, self-review log), 5.1 (domain model) and 5.2 (scenario tables and analysis sequence diagrams for UC-03 and UC-04).	devlog — commit 024c7af; docs/ai-use-log.md — commit fbcb528; docs/p2/diagrams — commit 1d2dfc5; docs/p2/sections — commit 0f7de31	#1
Abhishek Vemuri	Prototype evidence for Section 8: wrote the S-01 spike README (spikes/S-01-stripe/README.md) documenting the Stripe payment feasibility spike: the question, set-up, result and what remains untested.	PR #1: S-01 spike README, commit 6bec469; devlog entry, commit (the hash from step 3)	
Logan Tavares	Added section 6.1 architectural constraints paragraph, 6.2 choice explanation paragraph. Reviewed and corrected some minor grammar and spelling errors in sections 6.1, 6.2, and 7. Completed the 7.2 trade-off summary section. Reviewed the team's GitHub repository devlog to check project updates and documentation	11/8/2026 10:30 PM	

0.2 Changes since the P1 baseline

Change ID	What changed (IDs affected)	Why (P1 feedback, new finding, spike result)	Knock-on effect	Agreed on
CH-01	Added UC-07 Confirm Receipt and separated buyer confirmation from UC-04	Each use case should have one primary actor	Updated use-case analysis and handoff models	"Pending team confirmation — 11/08/2026"
CH-02	Updated textbook listing details to include course and price	UC-01 revealed missing listing attributes	Updated domain model and listing requirements	"Pending team confirmation — 11/08/2026"
CH-03	Tested Stripe sandbox payment feasibility	Needed evidence that third-party payments were possible	Supports the payment architecture; full integration remains untested	Pending team confirmation — 11/08/2026

1 to 3 Introduction, Overall Description and Requirements (carried forward)

1 Introduction

1.1 Purpose of this document and of the system

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This document is the updated requirements and design specification for BookBridge, created for CS 3354 Project Deliverable P2. It is written for the course instructor and teaching staff who will grade it, as well as the seven team members who will use it as a guide for building the system in P3. The document explains the problem, project scope, stakeholders, requirements, and design decisions the team has agreed on.

BookBridge addresses the difficulty students have when buying and selling used course textbooks with each other on campus. Students looking for used textbooks often have to search through social media groups or rely on word of mouth to find sellers. Students who have finished a course also need an easier way to find buyers for their books. BookBridge aims to solve this problem by providing a campus-based

platform where students can list textbooks, search for books they need, reserve them, and pay through a third-party payment service. Buyers and sellers will then arrange to meet on campus to complete the exchange.

1.2 Scope

In scope — we will build this	Out of scope — we will not, and why
Student accounts with sign-in using UTD credentials	Storing or processing card numbers (handled entirely by the payment processor; legal and security burden too large for this project)
Creating, editing and removing textbook listings (course, edition, condition, price)	Shipping or delivery (handoff is in person on campus only)
Searching and filtering listings by course, title or ISBN	Refund and dispute resolution workflows (handled outside the system by the team or the processor's own tools)
Reserving a listing and paying for it in the app through a third-party payment processor	Selling items other than course textbooks (the general-goods market is already served by the Discord group)

Why the exclusions were excluded: Card-number handling and refund and dispute workflows carry security, legal and operational obligations that a seven-person team cannot responsibly take on in one semester, so payment is delegated to an external processor and disputes are handled outside the system. Shipping and non-textbook goods are not needed to complete a textbook trade on campus and would widen the scope beyond what team can specify and later build, and general goods are already served by the existing Discord group.

1.3 Definitions, acronyms and abbreviations

Term or abbreviation	Meaning as used in this project
Listing	A seller's offer of one physical copy of a course textbook, with course, edition, condition and price
ISBN	International Standard Book Number; the identifier of a specific edition of a book, used to search and match listings
Available	Listing status: the book is listed and no buyer has paid for it
Reserved	Listing status: a buyer has paid for the book, and the handoff has not yet been confirmed
Sold	Listing status: the handoff has been confirmed by both parties and the listing is closed

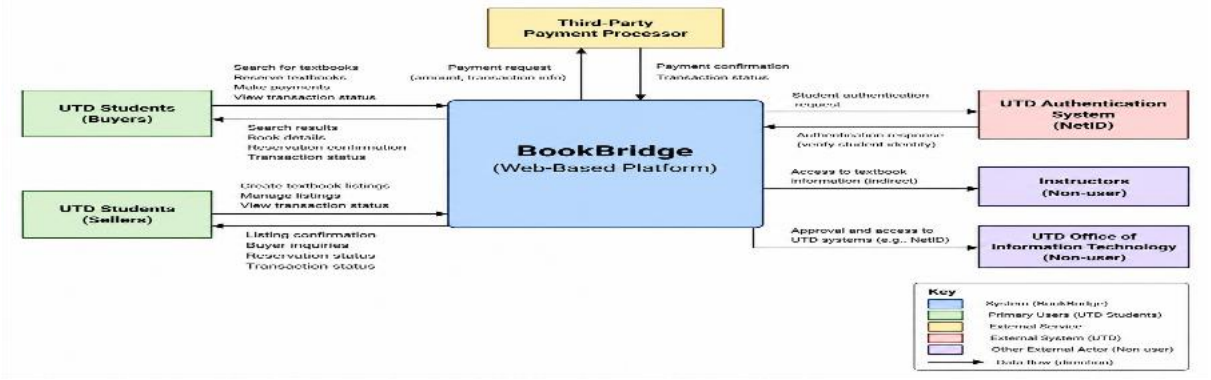
Handoff	The in-person exchange of the book between seller and buyer on campus
Payment processor	An external third-party service (for example, Stripe or PayPal) that takes the buyer's payment, so BookBridge never sees or stores card numbers
UTD credentials	University-issued login credentials used by students to access UTD services. BookBridge will use an authorized university authentication service to verify users.

1.4 References

[R1]	CS 3354 Fall 2026 Syllabus (Coordinator-aligned), The University of Texas at Dallas
[R2]	CS 3354 P1 Proposal and Requirements Team Template, Fall 2026
[R3]	P1 assignment page, Canvas (due 18 October 2026, 11:59 PM CT)
[R4]	ISO/IEC/IEEE 29148, *Systems and software engineering: Life cycle processes: Requirements engineering*

2 Overall Description

2.1 Product perspective and context



BookBridge is a web-based platform designed specifically for students at the University of Texas at Dallas (UTD) to buy and sell used textbooks with each other. Only verified UTD students will be able to access the platform. Students can create textbook listings, search for books by course, title, or ISBN, reserve textbooks, and make payments through a third-party payment processor. BookBridge will manage textbook listings, searches, reservations, and transaction status, while UTD's authentication system will verify student

identities, and the payment provider will process payments securely. The system will not handle shipping or delivery because all textbook exchanges will take place in person on or near the UTD campus.

2.2 Stakeholders and how you reached them

Stakeholder	Role in relation to the system	What they need from it	How you learned this
UTD Students buying books	Customers	Affordable textbooks	We assumed
UTD Students selling books	Seller	Sell used textbooks	We assumed
UTD administration	Potential Regulator	Safe transactions	We assumed
Instructors	Non-user	Correct textbook editions	We assumed
UTD OIT	Technical stakeholder	Secure UTD account verification	We assumed

2.3 User characteristics

Student Buyers

Student buyers are UTD students looking for affordable used textbooks for their classes. Most students are familiar with shopping online, but some may not have experience using a student-to-student marketplace. They need a simple way to search for textbooks by course, title, or ISBN, compare prices, and reserve books. Since many students will use BookBridge on their phones between classes, the platform should be mobile-friendly, easy to navigate, and accessible to students with disabilities.

Student Sellers

Student sellers are UTD students who want to sell textbooks they no longer need. They may have experience posting items for sale online, but the system should not assume that everyone knows how to create or manage listings. Sellers need an easy way to upload textbook information, set prices, update availability, and view reservations. The process should be quick and straightforward since students may be balancing classes, work, and other responsibilities.

2.4 Constraints

1. UTD Students Only – BookBridge will only allow students with valid UTD accounts to access the platform. Integration with UTD authentication will require university approval.

2. Third-Party Payments – All payments will be processed through an outside payment provider. BookBridge will not store credit card information or handle payment disputes directly.

3. Campus Pickup Only – Buyers and sellers must exchange textbooks in person on or near the UTD campus. Shipping and delivery will not be offered.

4. Time and Resources – Development must stay within the CS 3354 course deadlines and the team's available resources.

5. Textbooks Only – BookBridge will only support buying and selling used textbooks, not other products.

6. University Policies and Security – BookBridge must follow applicable UTD policies and protect student information.

2.5 Assumptions and dependencies

#	Assumption or dependency	What breaks if it does not hold
A1	Student Interest (Assumption) – UTD students will be interested in buying and selling used textbooks through BookBridge.	If students are not interested, the platform may have very few listings or buyers, making it less useful.
A2	**University Authentication (Dependency):** BookBridge depends on access to UTD's authentication system to verify student credentials.	If UTD does not approve access, the team will not be able to implement the planned login feature and will need to revise the authentication requirement.
A3	Payment Processing (Dependency) – A third-party payment provider will be available to process textbook purchases.	If the payment service cannot be integrated, students will not be able to complete payments through BookBridge as planned.
A4	Campus Exchanges (Assumption) – Students will be willing to meet on or near campus to exchange textbooks.	If students are unwilling to meet, completed transactions may decrease because BookBridge does not provide delivery.
A5	University Policies (Dependency) – The planned operation of BookBridge will be permitted under applicable UTD policies.	If university policies restrict certain activities, the team may need to change the system's features or operating model.

2.6 Feasibility summary

Feasibility gate	Verdict	Evidence behind the verdict

Technical	Feasible	Most of the features, like posting textbooks, searching, and creating accounts, can be built using basic web development tools. UTD login and online payments may be more difficult and will need further testing.
Operational	Feasible with some concerns	BookBridge would give UTD students another way to buy and sell used textbooks. However, we still need feedback from students to see if they would actually use it.
Schedule	Feasible	The project can be completed within the semester if we focus on the main features first. More complicated features may need to be added later depending on our progress.
Economic / resource	Feasible with limitations	We plan to use free development tools to keep costs low. Hosting and payment processing may have additional costs that we still need to look into.

3 Specific Requirements

3.1 Functional requirements

ID	Requirement — one "shall" sentence	Priority	Source	Acceptance criterion
FR-01	The system shall require users to authenticate through UTD's authorized login system before posting, reserving, or confirming textbook transactions.	MUST	UTD administration (2.2); CON-01	A tester can confirm this by trying to post, reserve and confirm while signed out; all three attempts are sent to NetID sign-in.
FR-02	The system shall allow a seller to list textbooks with the course, title, edition, ISBN, condition, and price.	MUST	UTD Students with books (2.2)	A tester can confirm this by posting a listing with all six required fields and finding it in search; leaving any required field empty blocks posting.
FR-03	The system shall allow a seller to edit or delete their listing at any time	MUST	UTD Students with books (2.2)	A tester can confirm this by changing one listing's price and deleting another; search shows the new price and no longer shows the deleted listing.

FR-04	The system shall allow buyers to search for available listings by course, title, or ISBN	MUST	UTD Students without books (2.2)	A tester can confirm this by searching seeded listings by course number, by title and by ISBN; each search returns every matching listing.
FR-05	The system shall allow buyers to filter search results by price and condition	COULD	UTD Students without books (2.2)	A tester can confirm this by setting a maximum price and a condition; only listings that meet both appear.
FR-06	The system shall allow a buyer to reserve an Available listing by completing payment through the external payment processor.	MUST	UTD Students without books (2.2); scope 1.2	A tester can confirm this by paying for an Available listing with a processor test card; the listing changes to Reserved under that buyer.
FR-07	The system shall change a listing from Reserved to Sold only after both the buyer and the seller confirm the handoff.	MUST	UTD Students with and without books (2.2); Sold definition (1.3)	A tester can confirm this by having only the seller confirm (listing stays Reserved), then the buyer confirm (listing shows Sold).
FR-08	The system shall return a listing to Available if the buyer does not complete payment within 10 minutes of selecting Reserve.	SHOULD	UTD Students with books (2.2)	A tester can confirm this by starting a reservation and abandoning payment; after 10 minutes the listing is Available again.
FR-09	The system shall notify the seller within 1 minute when one of their listings is reserved.	SHOULD	UTD Students with books (2.2)	A tester can confirm this by reserving a listing; the seller's account shows a notification within 1 minute.
FR-10	The system shall show only Available listings in buyer search results.	SHOULD	UTD Students without books (2.2)	A tester can confirm this by seeding listings in all three statuses; a search returns only the Available ones.
FR-11	The system shall arrange shipping or mail delivery of textbooks.	WON'T	Team scope decision (1.2, 2.4)	Agreed out of this release: handoff is in-person campus pickup only. A tester can confirm no shipping option

				appears anywhere in the system.

3.2 Quality requirements

ID	Requirement — with threshold, condition and method	Priority	Source	How it will be verified
QR-01	When a student searches listings by course number, title or ISBN, 95% of searches shall display results within 2 seconds, measured over one week of normal use with up to 100 concurrent users and 5,000 active listings.	SHOULD	UTD Students, buyers (2.2)	Load test: 100 simulated concurrent users run searches against 5,000 seeded listings; pass if the 95th-percentile response time is 2 seconds or less.
QR-02	A first-time seller shall be able to publish a complete listing (course, edition, condition, price) within 3 minutes without help, with at least 4 of 5 participants succeeding in a usability test.	SHOULD	UTD Students, sellers (2.2)	Moderated usability test with 5 UTD students who have never used BookBridge; time each from sign-in to published listing; pass if 4 of 5 finish within 3 minutes with no help.
QR-03	No buyer card number shall be stored in the BookBridge database or written to any BookBridge log file.	MUST	Scope decision (1.2): card handling is out of scope	Inspection: complete 20 test purchases using the payment processor's test card numbers, then search the database and all log files for those numbers; pass if there are zero matches.
QR-04	Every buyer and seller screen shall meet WCAG 2.1 Level AA, with listing status (Available, Reserved, Sold) never shown by color alone.	MUST	External: WCAG 2.1 Level AA	Automated accessibility audit of every screen with zero Level AA violations, plus a manual screen-reader walkthrough of search, reserve and pay; pass if both succeed.
QR-05	When 20 buyers try to reserve the same Available listing at the same moment, exactly one reservation shall succeed, in 100% of test runs.	MUST	UTD Students, buyers (2.2)	Automated concurrency test: send 20 simultaneous reservation requests for one listing, repeated 50 times; pass if every run ends with exactly one Reserved record.

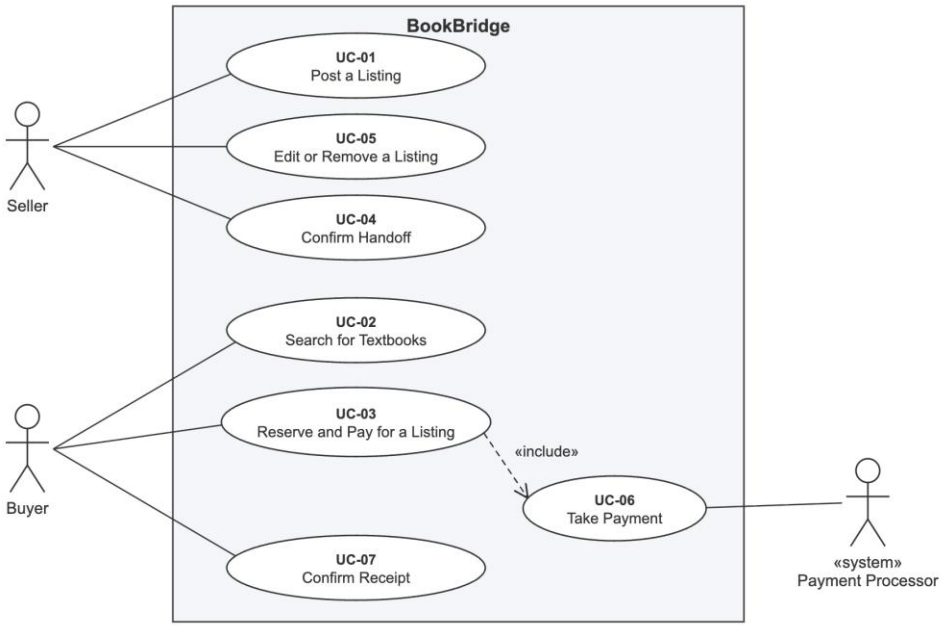
QR-06	BookBridge shall be available at least 99% of the time between 7:00 AM and 11:00 PM Central during the fall and spring semesters, measured monthly.	SHOULD	UTD Students (2.2)	External uptime monitor checks the site every 5 minutes; pass if the monthly report shows 99% or more successful checks inside the 7 AM to 11 PM window.

3.3 Constraint requirements

ID	Constraint — one sentence	Source	Why it cannot be negotiated
CON-01	BookBridge shall authenticate users only through UT Dallas's existing NetID single sign-on service.	UTD Office of Information Technology (owns NetID login); scope 1.2	Only the university's own login can prove a user is a current UTD student (see 'UTD credentials' in 1.3). The university controls student identity, not the team, so building our own account system is not an option.
CON-02	All card payments shall be processed by an external PCI DSS-compliant payment processor, so card data never reaches BookBridge's servers.	External: PCI Data Security Standard; scope 1.2	Card networks require PCI DSS compliance from anyone who stores or processes card data. A seven-person team cannot certify a system built in one semester, so card handling must stay with a certified processor.
CON-03	A working version of BookBridge shall be demonstrated by the CS 3354 P3 deadline in December 2026.	CS 3354 instructor: syllabus and P3 assignment	The instructor sets milestone dates and the syllabus states that late submissions are not accepted. Scope must shrink to fit the date; the date cannot move to fit the scope.

3.4 Initial use cases

Changed since v0.3: the full expansions of UC-03 and UC-04 moved to 4.2, and UC-07 Confirm Receipt was added when UC-04 was split so that each use case has one actor (change in 0.2). Signing in through UTD NetID is still a precondition, not a use case (FR-01, CON-01).



Use case	Primary actor	TUCBW (begins when)	TUCEW (ends when)
UC-01 Post a Listing	Seller	The Seller asks to post a new textbook listing.	The Seller sees the new listing in their listings with the status Available.
UC-02 Search for Textbooks	Buyer	The Buyer searches by course number, title or ISBN.	The Buyer sees the matching Available listings, or that none match.
UC-03 Reserve and Pay for a Listing	Buyer	The Buyer asks to reserve an Available listing they found in search.	The Buyer sees that the listing is Reserved for them, with the price paid and the Seller's UTD email.
UC-04 Confirm Handoff	Seller	The Seller reports that the book for one of their Reserved listings has been handed over.	The Seller sees their confirmation recorded.
UC-05 Edit or Remove a Listing	Seller	The Seller chooses one of their listings to change or remove.	The Seller sees the updated listing, or that it is no longer listed.
UC-06 Take Payment (included by UC-03)	Buyer, with the Payment Processor supporting	UC-03 passes the Buyer to the Payment Processor with the amount due.	The Payment Processor reports whether the payment succeeded.
UC-07 Confirm Receipt	Buyer	The Buyer reports that they received the book for a Reserved listing.	The Buyer sees their confirmation recorded, and the listing shown as Sold once the Seller has confirmed.

UC-01, UC-03 and UC-04 are expanded in 4.2. UC-02, UC-05, UC-06 and UC-07 stay at this level (see 9.2).

3.5 Requirements to use case traceability

Replaced by the extended traceability matrix in 9.1.

4 Use-Case Analysis

Chapter 7, first half. Here you take the use cases from section 3.4 and write the most important of them out turn by turn, still with the technology switched off.

4.1 Use cases selected for analysis

Use case	Name	Why it was selected	Non-trivial steps
UC-03	Reserve and Pay for a Listing	Specified in full at P1. It carries most of the product, it is the only use case that involves the Payment Processor, and it is the one we are least sure we can build (R2). It touches the most quality requirements (QR-03, QR-05) and has a failure the Buyer can see (declined payment).	(2), (4), (6)
UC-04	Confirm Handoff	Specified in full at P1. It is the only way a listing becomes Sold (FR-07), and it needs both the Seller and the Buyer to confirm, which is easy to get wrong.	(0), (4)
UC-01	Post a Listing	Every trade starts with a listing, and QR-02 depends on it. Expanding it showed that FR-02 and the Listing definition in 1.3 disagree about which items a listing has.	(0), (4), (6)

4.2 Expanded use cases

UC-03 Reserve and Pay for a Listing

Primary actor	Buyer

Supporting actors and external systems	Payment Processor (external system that takes the payment, so card details never reach BookBridge; CON-02)
TUCBW (this use case begins when)	The Buyer asks to reserve an Available listing they found in search (UC-02).
TUCEW (this use case ends when)	The Buyer sees that the listing is Reserved for them, with the price paid and the Seller's UTD email address to arrange the handoff.
Preconditions	The Buyer is signed in through UTD NetID (FR-01, CON-01). The Buyer has opened a listing from the search results (UC-02).
Postconditions	The listing is Reserved, linked to this Buyer, and no longer appears in buyer search results (FR-10). The Payment Processor holds a record of the completed payment. A reservation notice is waiting for the Seller (FR-09). BookBridge has stored no card details (QR-03).
Traces to requirements	FR-01, FR-06, FR-08, FR-09, FR-10; QR-03, QR-04, QR-05; CON-01, CON-02

Actor input / action	System response
	(0) The system shows the listing the Buyer opened: course, title, edition, ISBN, condition, price, the Seller's name and the listing status (Available).
(1) The Buyer asks to reserve the listing.	(2) The system holds the listing for the Buyer and shows the price to pay, the time left on the 10-minute hold, and a request to go ahead with payment.
(3) The Buyer agrees to pay.	(4) The system passes the Buyer to the Payment Processor with the amount due.
(5) The Buyer gives their payment details to the Payment Processor.	(6) The system shows the Buyer that the listing is now Reserved for them, with the title, the price paid, the Seller's name and the Seller's UTD email address to arrange the handoff.
(7) The Buyer reads the reservation confirmation.	

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	(1)	The listing belongs to the Buyer.	The system tells the Buyer they cannot reserve their own listing.	Ends.
A2	(2)	Another Buyer has already held or reserved the listing (QR-05).	The system tells the Buyer the listing is no longer available and shows the search results again.	Ends.

A3	(3)	The Buyer decides not to pay.	The system releases the hold and shows the listing as Available again.	Ends.
A4	(5)	The Buyer does not finish paying within 10 minutes (FR-08).	The system releases the hold, and the listing returns to Available. When the Buyer comes back, the system tells them the hold ended and the listing was not reserved.	Ends.
A5	(5)	The Payment Processor declines the payment.	The system tells the Buyer the payment was declined, keeps the hold for the time left, and asks them to try another payment method.	Rejoins at (3).
A6	(5)	The Payment Processor does not report a result.	The system tells the Buyer the payment could not be confirmed and the listing is not reserved yet. The hold stays until it ends (A4).	Ends.

UC-04 Confirm Handoff

Primary actor	Seller
Supporting actors and external systems	None. The Buyer confirms receipt separately in UC-07 Confirm Receipt.
TUCBW (this use case begins when)	The Seller reports that the book for one of their Reserved listings has been handed to the Buyer.
TUCEW (this use case ends when)	The Seller sees their handoff confirmation recorded and the listing shown as waiting for the Buyer to confirm receipt. This use case continues with UC-07 Confirm Receipt (Buyer), which closes the trade.
Preconditions	The Seller is signed in through UTD NetID (FR-01, CON-01).
Postconditions	The Seller's confirmation is recorded against the listing with its time. The listing stays Reserved until the Buyer confirms receipt (UC-07), and then becomes Sold (FR-07).
Traces to requirements	FR-01, FR-07; QR-04; CON-01

Actor input / action	System response
	(0) The system shows the Seller's listings with the status of each. Each Reserved listing also shows the Buyer's name, the date it was reserved and whether the Buyer has confirmed receipt.
(1) The Seller reports that the book for one Reserved listing has been handed over.	(2) The system shows the listing's title, the Buyer's name and the price paid, and asks the Seller to confirm that the book changed hands.
(3) The Seller confirms.	(4) The system shows that the Seller's confirmation is recorded with its time, and that the listing stays Reserved until the Buyer confirms receipt.
(5) The Seller sees the confirmation.	

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	(0)	The Seller has no Reserved listings.	The system shows that there is no handoff to confirm.	Ends.
A2	(3)	The Seller says the book has not changed hands yet.	The system records nothing and the listing stays Reserved.	Ends.
A3	(3)	The Seller reports that the Buyer did not come to the handoff.	The system keeps the listing Reserved and directs the Seller to dispute handling outside BookBridge (out of scope, 1.2).	Ends.
A4	(4)	The Buyer has already confirmed receipt (UC-07).	The system marks the listing Sold, removes it from search results, and shows the Seller that the trade is complete (FR-07).	Ends.

UC-01 Post a Listing

Primary actor	Seller
Supporting actors and external systems	None
TUCBW (this use case begins when)	The Seller asks to post a new textbook listing.
TUCEW (this use case ends when)	The Seller sees the new listing in their listings with the status Available.
Preconditions	The Seller is signed in through UTD NetID (FR-01, CON-01).
Postconditions	A listing with the course, title, edition, ISBN, condition and price is recorded under this Seller with the status Available, and it appears in buyer search results (FR-04, FR-10).
Traces to requirements	FR-01, FR-02, FR-04, FR-10; QR-02, QR-04; CON-01

Actor input / action	System response
	(0) The system shows the Seller's listings, each with its title, price and status, and offers to post a new listing. If the Seller has no listings, it says so.
(1) The Seller asks to post a new listing.	(2) The system asks for the course, title, edition, ISBN, condition and price, and shows the condition grades a listing can have.
(3) The Seller enters the course, title, edition, ISBN, condition and price.	(4) The system shows the listing as Buyers will see it and asks the Seller to confirm.
(5) The Seller confirms.	(6) The system shows the new listing in the Seller's listings with the status Available.
(7) The Seller sees the new listing.	

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	(3)	A required item is missing, or the price	The system names the item to fix and keeps everything else the Seller entered (FR-02).	Rejoins at (3).

		is not an amount greater than zero.		
A2	(3)	The ISBN is not a valid 10- or 13-digit ISBN.	The system says the ISBN is not valid and asks the Seller to check it.	Rejoins at (3).
A3	(5)	The Seller decides not to post.	The system posts nothing and shows the Seller's listings unchanged.	Ends.

4.3 Self-review log

Use case	Check from the Lesson 13 review checklist	Defect found	How it was fixed
UC-03	2. Is there a step (0)?	P1 had none, and nothing told the Buyer how to reach the Seller after paying.	Added step (0). Step (6) now shows the Seller's UTD email (change in 0.2).
UC-03	3. Does every row hold one turn?	P1 step 4 placed the hold and sent the Buyer to payment in one row.	Split into steps (2) and (4).
UC-03, UC-04	6. Are preconditions things you may assume?	"The listing is Available" and "the Buyer is not the Seller" (UC-03), and "the book has changed hands" (UC-04), can all be false.	Moved to UC-03 A1 and A2, and UC-04 A2 and A3.
UC-03, UC-04	9. Any background processing?	UC-03 step 8 notified the Seller, and UC-04 3a sent a 7-day reminder. Neither actor sees or waits for them.	The Seller notice is now a UC-03 postcondition (FR-09). The 7-day case could not be decided; added to section 11.
UC-03, UC-04	10. Does any row name a technology or screen?	"Payment page" (UC-03 step 4) and "Mark as Handed Off" (UC-04 step 1).	Reworded without the page or the control.
UC-03	Alternate-flow sweep ("what if this does not work?")	P1 6a released the hold on a decline, so the Buyer could not retry. P1 6b said no charge was made,	A5 keeps the hold; A6 says the payment could not be confirmed. A payment after the hold ends

		which the system cannot know.	could not be decided; added to section 11.
UC-04	One actor, one goal (TUCCW)	Both the Seller and the Buyer acted in one use case.	The Seller is the only actor; the Buyer's side continues in UC-07 Confirm Receipt (change in 0.2).
UC-01	4. Is every item the actor supplies a domain model attribute?	Course and price are supplied but missing from FR-02, and no condition grades are defined.	Course and price added to 5.1 and FR-02 (change in 0.2). Grades could not be decided; added to section 11.
All three	1, 5, 7, 8	None found.	No change needed.

5 Analysis Models

Chapter 7, second half. What has to exist inside the system for the use cases in section 4 to happen, and how those parts talk to each other.

5.1 Domain model

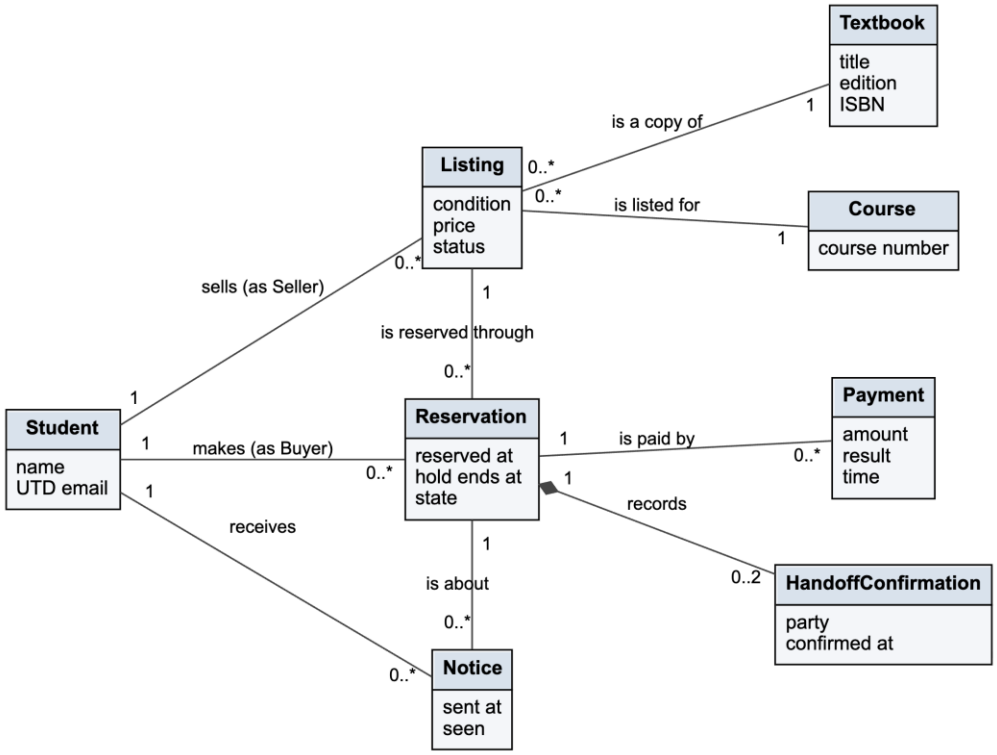


Figure 5.1 Domain model (conceptual classes, attributes, associations with multiplicities)

Class	What it represents (one line)	Key attributes	Associations (with multiplicity)
Student	A UTD student signed in through NetID. The same student can be the Seller of one listing and the Buyer of another.	Name, UTD email	Sells 0..* Listing; makes 0..* Reservation; receives 0..* Notice
Listing	A Seller's offer of one physical copy of a course textbook (see 'Listing' in 1.3).	Condition, price, status (Available, Reserved, Sold)	Sold by 1 Student; is a copy of 1 Textbook; is listed for 1 Course; is reserved through 0..* Reservation
Textbook	One edition of a book, identified by its ISBN. This is what a Buyer searches for.	Title, edition, ISBN	Has 0..* Listing
Course	A UTD course that a listing is offered for, so	Course number	Has 0..* Listing

	Buyers can search by course (FR-04).		
Reservation	A Buyer's 10-minute hold on a listing, which becomes a paid reservation when payment succeeds (FR-06, FR-08).	Reserved at, hold ends at, state (Held, Paid, Released)	Made by 1 Student; for 1 Listing; is paid by 0..* Payment; records 0..2 HandoffConfirmation (composition); is the subject of 0..* Notice
Payment	BookBridge's record of one payment attempt reported by the Payment Processor. It never holds card details (QR-03).	Amount, result (Succeeded, Declined, No answer), time	For 1 Reservation
HandoffConfirmation	One party's confirmation that the book changed hands. The listing is Sold once both exist (FR-07).	Party (Seller or Buyer), confirmed at	Part of 1 Reservation
Notice	A message the system leaves for a student, such as a new reservation for the Seller (FR-09).	Sent at, seen	Received by 1 Student; is about 1 Reservation

What the model revealed: the word "book" was doing two jobs. A Textbook is an edition, which a Buyer searches for by title or ISBN (FR-04). A Listing is the one physical copy a Seller offers, and only the Listing has a condition, a price and a status (see 'Listing' in 1.3). Many listings can be copies of the same textbook, so we modelled them as two classes rather than one.

What stays outside: the Payment Processor and UTD NetID are external systems, not classes (CON-01, CON-02). Payment is BookBridge's own record of the result the Payment Processor reports, so there is nowhere in the model to put a card number (QR-03).

What must always be true: a listing has at most one Reservation in the Held or Paid state at a time (QR-05). A Paid reservation is what makes the listing Reserved, and two HandoffConfirmations, one from the Seller and one from the Buyer, are what make it Sold (FR-07).

5.2 Object interaction models

5.2.1 UC-03 Reserve and Pay for a Listing

(a) Non-trivial steps. Steps (2), (4) and (6) need objects inside the system. Step (2) checks that the listing is still free and places the hold, step (4) asks the Payment Processor for payment, and step

(6) records the payment, reserves the listing, leaves the Seller a notice and fetches the Seller's contact details.

The other steps are trivial. Step (0) shows the listing the Buyer already opened in UC-02. Steps (1) and (3) are the Buyer's inputs, so they appear only as the messages that start the work. Step (5) happens between the Buyer and the Payment Processor, outside BookBridge. Step (7) is the Buyer reading the confirmation.

ReservationDesk is an interaction object: the part of BookBridge that runs one reservation for the Buyer, and it is in control of the whole interaction. The Payment Processor is an external system outside the boundary (CON-02). Every other object is a class from 5.1.

(b) Scenario table.

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)
1	Buyer	Asks to reserve the listing	The listing	ReservationDesk
2a	ReservationDesk	Asks whether the listing is Available and not held	The listing	Listing
2b	ReservationDesk	Creates a 10-minute hold for the Buyer	The Buyer, hold ends at	Reservation
2c	ReservationDesk	Shows the price and the time left on the hold	Price, hold ends at	Buyer
3	Buyer	Agrees to pay	None	ReservationDesk
4	ReservationDesk	Asks for payment of the amount due	Amount	Payment Processor
6a	Payment Processor	Reports that the payment succeeded	Amount, result	ReservationDesk
6b	ReservationDesk	Records the payment	Amount, result, time	Payment
6c	ReservationDesk	Marks the reservation Paid	State	Reservation
6d	ReservationDesk	Changes the listing status to Reserved	Status	Listing
6e	ReservationDesk	Leaves a reservation notice for the Seller	The reservation	Notice
6f	ReservationDesk	Asks for the Seller's name and UTD email	The Seller	Student (Seller)

6g	ReservationDesk	Shows the reservation confirmation	Title, price paid, Seller's name and UTD email	Buyer

(c) Analysis sequence diagram.

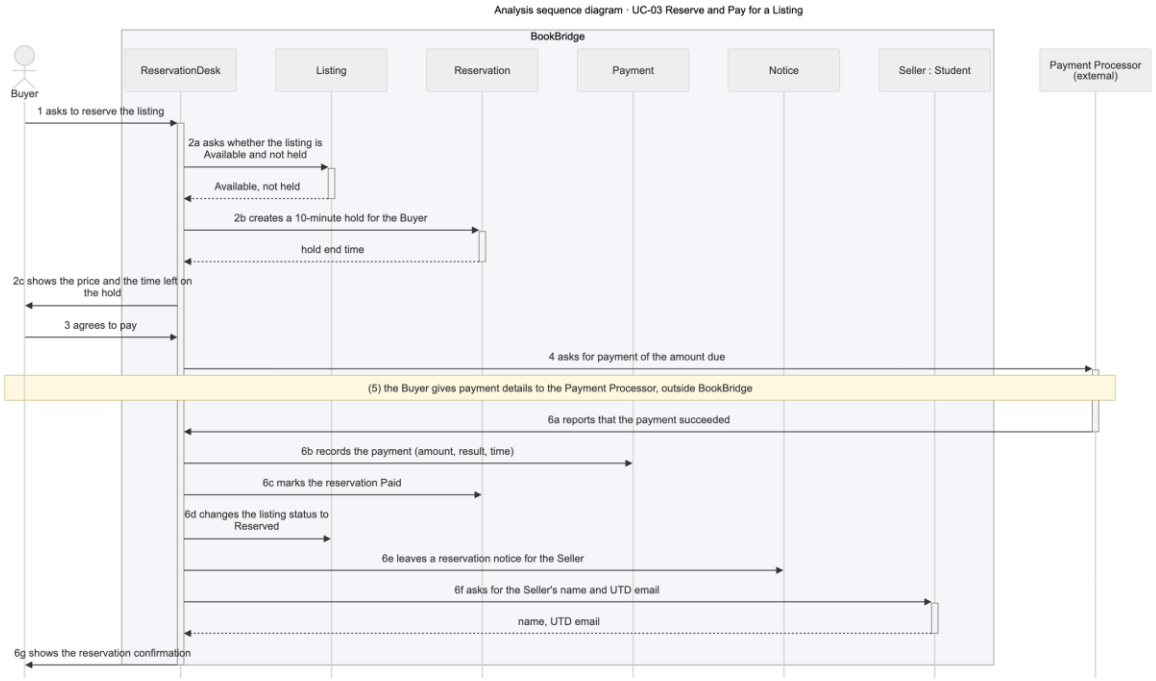


Figure 5.2.1 Analysis sequence diagram for UC-03; message numbers match the scenario table

5.2.2 UC-04 Confirm Handoff

(a) Non-trivial steps. Steps (0) and (4) need objects inside the system. Step (0) fetches the Seller's listings and, for each Reserved one, the Buyer, the date reserved and whether the Buyer has confirmed. Step (4) records the Seller's confirmation.

The other steps are trivial. Steps (1) and (3) are the Seller's inputs, so they appear only as the messages that start the work. Step (2) shows details the system already fetched at step (0). Step (5) is the Seller reading the confirmation.

HandoffDesk is an interaction object: the part of BookBridge that runs a handoff confirmation for the Seller, and it is in control of the whole interaction. Every other object is a class from 5.1.

(b) Scenario table.

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)
0a	HandoffDesk	Asks for the Seller's listings and their status	The Seller	Listing
0b	HandoffDesk	Asks for the Buyer, date reserved and	The Reserved listings	Reservation

		confirmations of each Reserved listing		
0c	HandoffDesk	Shows the listings with their Buyer details	Status, Buyer's name, date reserved, whether the Buyer has confirmed	Seller
1	Seller	Reports that the book for one Reserved listing was handed over	The listing	HandoffDesk
2	HandoffDesk	Asks the Seller to confirm the handoff	Title, Buyer's name, price paid	Seller
3	Seller	Confirms the handoff	None	HandoffDesk
4a	HandoffDesk	Asks the reservation to record the Seller's confirmation	Party (Seller)	Reservation
4b	Reservation	Records the confirmation with its time	Party, confirmed at	HandoffConfirmation
4c	HandoffDesk	Shows that the confirmation is recorded	Confirmed at, listing stays Reserved	Seller

(c) Analysis sequence diagram.

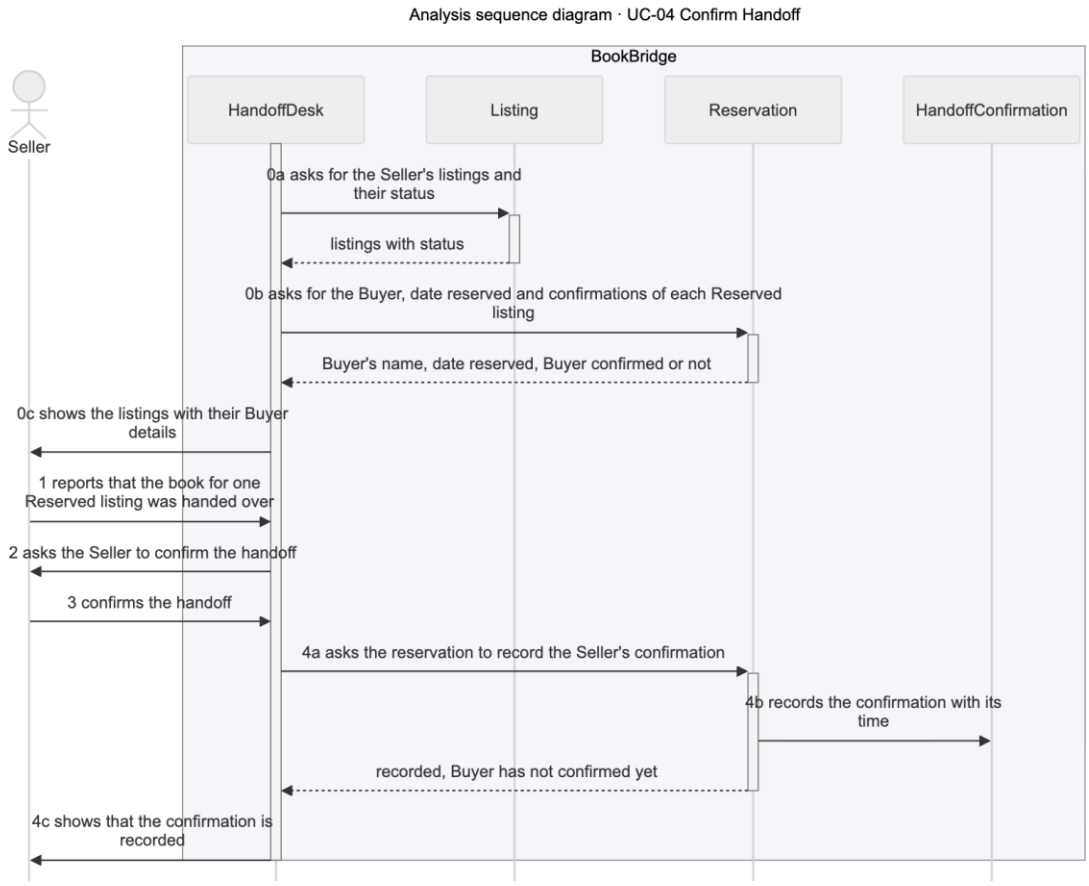


Figure 5.2.2 Analysis sequence diagram for UC-04; message numbers match the scenario table

5.3 Consistency cross-check

Check	Result (pass, or what failed)	Fixed in
K1 Every item in an actor column (4.2) is an attribute in the domain model (5.1)	Pass: All actor inputs in UC-01, 03, 04 exist in 5.1, and card details in UC-03 are excluded (QR-03 and CON-02).	No fix needed
K2 Every object on a sequence diagram is a domain class or a declared interaction object	Pass: In 5.2.1, lifelines are domain classes, declared interaction object ReservationDesk, and external PaymentProcessor. In 5.2.2 (UC-04), lifelines are domain classes and declared interaction object HandoffDesk	No fix needed
K3 Every message on a sequence diagram corresponds to exactly one scenario-table row	Pass: All 13 messages in 5.2.1 and all 9 messages in 5.2.2 match their scenario table rows in 5.2 (b).	None needed
K4 Every non-trivial step in 5.2 (a) has at least one scenario-table row	Pass: UC-03 non-trivial steps 2, 4, 6 are realized by rows 2a-2c, 4, and 6a-6g. UC-04 non-trivial steps 0 and 4 are realized by rows 0a-0c and 4a-4c.	None needed
K5 Every analyzed use case traces to at least one functional requirement	Pass: UC-01 traces to FR-01, 02, 04, 10, UC-03 races to FR-01, 06, 08, 09, 10, UC-04 traces to FR-01, 07.	No fix needed
K6 The last messages of each diagram make the use case postcondition true	Pass: In 5.2.1, messages 6b-6e	None needed

	recorded the Payment, mark the Reservation Paid, set Listing status to Reserved, and create a Notice for the seller without storing car data. In 5.2.2, messages 4a-4b record the seller's HandoffConfirmation with its timestamp while keeping the listing Reserved until UC-07.	

6 High-Level Design (Architecture)

Chapter 8. The rule now reverses: a high-level design that does not say where things run, what they are built from and what that choice cost has not been done.

6.1 Architectural drivers: quality-attribute scenarios

QAS	Quality attribute	Stimulus	Artifact	Response	Response measure	From
QAS-01	performance	A student initiates a search request for listings using a course number, title, or ISBN.	The system's search engine/database lookup service.	The system processes the query and displays the matching search results to the student.	95% of searches must display results within 2 seconds, measured over one week of normal use under a load of up to 100 concurrent users and 5,000 active listings.	QR-01
QAS-02	usability	A first-time seller attempts to create and publish a	The listing creation and publishing subsystem	The user completes and publishes the listing successfully	At least 80% (4 out of 5 participants) successfully complete the	QR-02

		complete textbook listing containing specific, required attributes		without receiving external documentation or support help.	task during the usability test within 3 minutes.	
QAS-03	security	An attempt or transaction execution that transmits or processes a buyer's card number within BookBridge.	BookBridge core system	The database filters out or ignores card number fields during inserts.	0 instances of cleartext or raw buyer card numbers are found in any BookBridge database tables or system log files during automated security audits, static/dynamic code analysis, or database inspections.	QR-03
6.1 Architectural constraints	The architecture has three main constraints. The first CON-01 is that identity verification would be through the university's NetID Single Sign-On for authentication. CON-02 requires that all card payments must be processed through an external PCI DSS-compliant payment processor so that BookBridge doesn't store card information. CON-03 requires a working version of BookBridge to be available by the P3 deadline in December.					

6.2 Architectural style and alternatives

Style considered	Drivers it serves	What it costs	Verdict
Model-view-controller	Modifiability & Maintainability: Separates business logic from UI data presentation, isolating	Performance Overhead: Multiple layer boundaries introduce data transformation and method-call overhead.	Chosen

	changes to specific tiers.		
Pipe-and-filter	Stream Processing: Optimizes sequential data transformations.	Poor Interactive Experience: Extremely difficult to manage complex, bidirectional user-interaction state efficiently.	Rejected
Repository	Data Centralization: Simplifies uniform access to a single, shared data store.	Single Point of Failure: Tight coupling to the schema risks performance bottlenecks and reduces deployment flexibility.	Rejected
Choice explanation	The team selected the Model-View-Controller architectural style for BookBridge. This was selected to allow team members to work on different tasks independently. This will help the team meet the P3 deadline. Overall, this style will improve organization, maintainability, and division of responsibilities.		

6.3 Component view

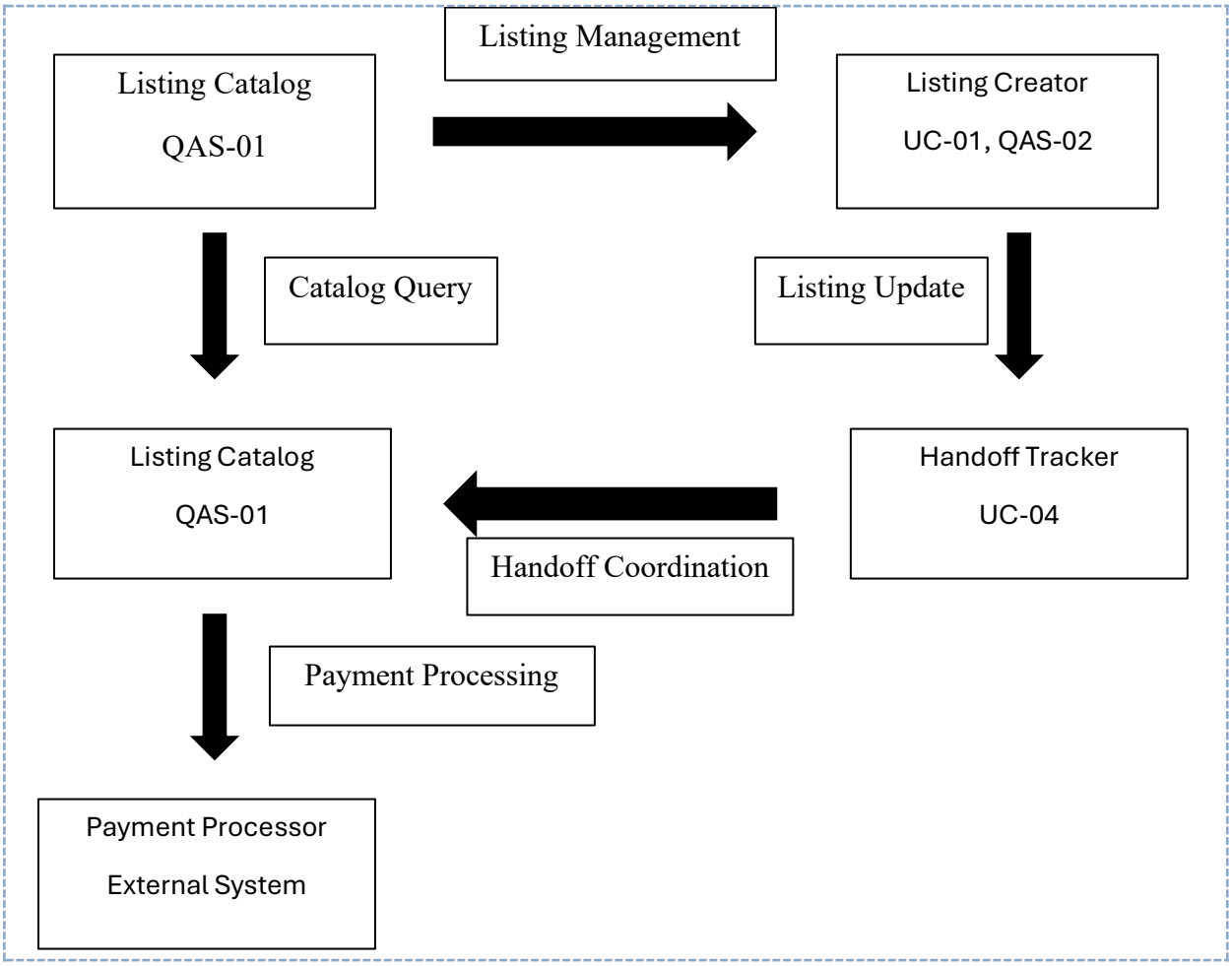


Figure 6.3 Component diagram: components, provided and required interfaces, external systems

Component	Responsibility (one sentence)	Provides	Requires	Answers to (UC / QAS / ADR)
Listing Catalog	Manages fast read-only index lookups and text matching for all active textbook offerings.	Catalog Query	None	QAS-01
Listing Creator	Guides sellers through creating listings with complete, validated attributes that align conflicting definitions.	Listing Management	Listing Update	UC-01 QAS-02
Order Manager	Handles transactional state changes for reserving items and secure out-of-scope payment handoffs.	Catalog Query, Listing Update	Payment Processing	UC-03 QAS-03

Handoff Tracker	Mediates and records bilateral fulfillment confirmations from both buyers and sellers before closing an order.	Handoff Coordination	Listing Update	UC-04

6.4 Deployment view

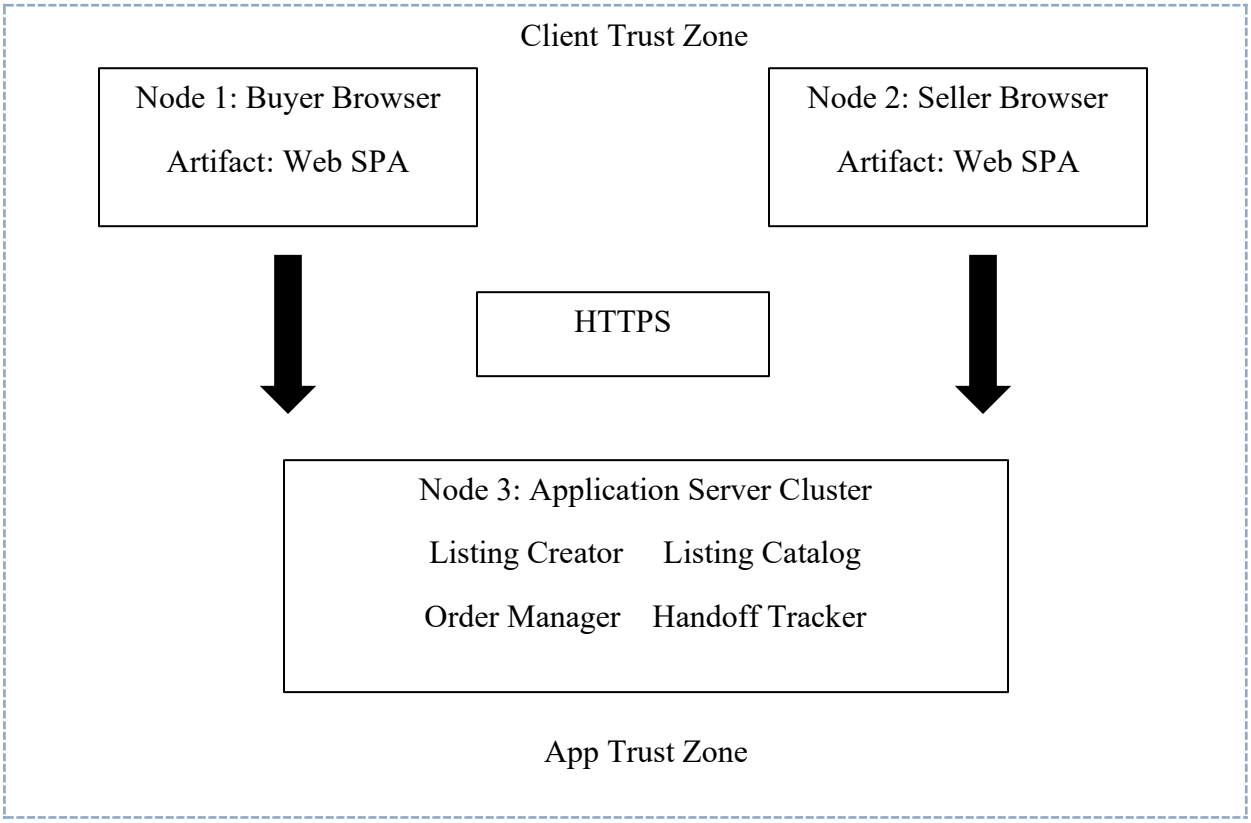


Figure 6.4 UML deployment diagram: nodes, execution environments, deployed artifacts, communication paths with protocols, trust boundaries

Node	What runs on it (artifacts, components)	Communicates with (protocol)	Trust zone
Buyer Browser	Web SPA frontend footprint asset	Node 3 via HTTPS	Client Trust Zone
Seller Browser	Web SPA frontend footprint asset	Node 3 via HTTPS	Client Trust Zone
Application Server Cluster	Listing Catalog, Listing Creator, Order Manager, Handoff Tracker	Nodes 1 & 2 via HTTPS	App Trust Zone

6.5 From analysis to architecture

Analysis object (from 5.1 and 5.2)	Component that owns it (6.3)	Note (split, merged, external)
Student	Shared across all components (net ID)	Split/external: Identity verified externally with NetID SSO (CON-01). Order manager and Listing Creator reference ID/email
Listing	Listing Creator, Listing Catalog, Order Manager	Split: Created in Listing Creator (UC-01), indexed in Listing Catalog (QAS-01), and status is updated (Available > Reserved > Sold) by Order Manager and Handoff Tracker
Textbook	Listing Catalog	Populated with Listing Creator (UC-01) and indexed by Listing Catalog for title/edition queries (QAS-01)
Course	Listing Catalog	Associated with listings on creation (Listing Creator) and queried in Listing Catalog (FR-04, QAS-01)
Reservation	Order Manager	Owned by Order Manager to manage the 10-minute hold and reservation state (Held, Paid, Released)
Payment	Order Manager	External/Merged: Card processing is handled by the external Payment Processor (CON-02, QR-03). Order Manager only records the processor's result (Succeeded, Declined, No answer)
Handoff Confirmation	Handoff Tracker	Owned by Handoff Tracker to record bilateral Seller (UC-04) and Buyer (UC-07) confirmations

Notice	Order Manager	Generated by Order Manager when a reservation is paid to notify the Seller within 1 minute (FR-09)
ReservationDesk	Order Manager	Merged: Interaction coordinator for UC-03 realized inside Order Manager
HandoffDesk	Handoff Tracker	Merged: Interaction coordinator for UC-04 realized inside Handoff Tracker

6.6 Design principles applied

Principle	Where it shows in your design	Evidence (component, interface, ADR)
Separation of concerns	Demonstrated by decoupling the fast read-optimized indexing onto its own dedicated query node away from write-heavy processes.	Listing Catalog
Information hiding	Enforced at Boundary 2, where financial details are hidden from our system by handing off transactions completely out-of-scope to a third-party tokenized interface.	Trust Boundary 2
Low coupling	Achieved by connecting components exclusively via network-isolated gRPC/HTTPS protocols, ensuring binary changes in one service do not break dependencies in another.	gRPC/HTTPS
High cohesion	the Handoff Tracker, which mediates and records bilateral fulfillment confirmations.	Handoff Tracker
Design for change	left blank because the static 3-node allocation focuses entirely on the architectural footprint without giving structural extensibility abstractions	

Keep it simple (KISS)	Maintained by avoiding complex distributed event streaming frameworks and utilizing direct, synchronous gRPC point-to-point routing between nodes.	gRPC

7 Design Rationale

7.1 Architecture decision records

At least three. ADR-01 records the style decision from 6.2. Use the format you practiced in the Lesson 16 workshop, written about your own project. Copy the block for each decision.

ADR-01 Use Model-View-Controller Architecture

Status	Accepted
Date and owner	November 8, 2026, Abran Garica
Context (drivers)	CON-03 needs a working demo by the P3 deadline. This structure allows our group to divide the work easier. QAS-01 needs search results to appear in 2 seconds, and QAS-02 needs first time sellers to publish a listing in 3 minutes. Keeping the UI separate from the business logic lets us rework each of these without changing the other.
Options considered	Model-View-Controller Pipe-and-filter Repository
Decision	One server application holds the models, controllers, and views back by one database.
Consequences	Quality: Using this architecture, screens can be reworked after testing without touching listing or reservation rules. Cost accepted: every request crosses the view, controller, and model layers, which adds some delay to search.
Evidence	None yet. The load test and usability test are planned for P3.
Related	ADR-, QAS-01, QAS-02, UC-01, UC-02

ADR-02 Payment through Stripe's hosted checkout page

Status	Accepted
Date and owner	November 8, 2026, Abran Garcia

Context (drivers)	QAS-03 says no card numbers are stored in our database or logs. CON-02 and UC-03 require card payments to be processed by an external payment processor.
Options considered	Redirect buyer to Stripe's checkout page Embed a card form on our own page Our own card form posting to our server
Decision	BookBridge sends the amount and listing to Stripe and receives teh result from Stripe. No card information is inputted to BookBridge.
Consequences	Quality: We don't collect any card data and have to deal with the extra security of storing that data. Cost accepted: Little control over what the payment page looks like. Dependency on Stripe being up.
Evidence	Spike S-01 shows a successful $25 sandbox payment through the hosted checkout
Related	ADR-, QAS-03, R-2, UC-03, S-01

ADR-03 Responsive web application instead of native mobile apps

Status	Accepted — November 8, 2026
Date and owner	November 8, 2026, Abran Garcia
Context (drivers)	QR-04 requires accessibility on every screen. CON-03 requires a working demo by P3, it is unreasonable to build and maintain several versions
Options considered	One web application that adapts to phones and laptos Separate native iOS and Android apps A website designed for laptops/computers only
Decision	Students can open BookBridge in their browser on any device
Consequences	Quality: Favors reach while also make it easier to develop. Cost accepted: No phone push notifications and less access to phone features, so the seller notification in FR-09 appears inside the seller's account rather t
Evidence	None yet. The building and accessibility audit for QR-04 is planned for P3
Related	ADR-1, QAS-2, FR-09

7.2 Trade-off summary

Quality favored	Architectural move	Cost accepted	Recorded in
Modifiability and Maintainability	Use Model-View-Controller to separate the user interface and data management.	Coordination between separation of responsibilities and extra time needed for the system to complete tasks.	ADR-01
Security	Using Stripe to keep card information outside of BookBridge's servers.	Less control over payment, and reliance on Stripe for availability and integrity.	ADR-02
Performance and Accessibility	Using a separate web application that allows students to access BookBridge via phone or computer without needing a separate application.	The system will have less device specific features and can't provide native notifications. Some aspects may also be less optimized than dedicated applications.	ADR-03 and QAS-01

8 Prototype Evidence: Architecture Spike

Spike ID and title	S-01— Stripe Payment Feasibility
Risk or decision it addresses	R2 — Payment Processor Integration
Question (answerable yes or no)	Can Stripe's hosted checkout accept a sandbox textbook payment without BookBridge collecting card details?
Set-up	Stripe sandbox, $25 Calculus Textbook, one-time payment, hosted checkout
How it was measured	One test checkout completed; transaction status checked in Stripe dashboard
Result (numbers)	1 successful test payment for $25; no real money transferred
What it changed or confirmed	Stripe hosted checkout is feasible for a prototype; actual BookBridge integration and webhook handling remain untested
Where to see it	`docs/payment-feasibility.md`, commit `af97f4c`

9 Traceability

9.1 Extended traceability matrix

Requirement	Use case	Analysis classes	Component(s)	QAS / ADR	Evidence
FR-01	Precondition of UC-01, UC-03, UC-04, UC-05, UC-07	Student	External: NetID SSO (all components)	CON-01	Open question Q3/Risk R1
FR-02	UC-01	Student, Listing, Textbook, Course	Listing Creator	QAS-02, ADR-01	4.2 UC-01 analysis
FR-03	UC-05	Student, Listing	Listing Creator	ADR-01	6.3 design
FR-04	UC-01, UC-02	Listing, Textbook, Course	Listing Catalog	QAS-01, ADR-01	6.3 design
FR-05	UC-02	Listing	Listing Catalog	QAS-01	6.3 design
FR-06	UC-03	Student, Listing, Reservation, Payment, ReservationDesk	Order Manager, External payment processor	QAS-03, CON-02	Spike S-01
FR-07	UC-04, UC-07	Student, Listing, Reservation, HandoffConfirmation, HandoffDesk	Handoff Tracker, Order Manager	ADR-01	5.2 sequence diagram
FR-08	UC-03 (A4)	Listing, Reservation, ReservationDesk	Order Manager	ADR-01	UC-03 alternate flow A4
FR-09	UC-03 (postcondition)	Student, Reservation, Notice, ReservationDesk	Order Manager	ADR-01	UC-03 postconditions
FR-10	UC-01, UC-02, UC-03	Listing	Listing Catalog,	QAS-01	UC-01, UC-03 postconditions

			Order Manager		
FR-11	None (won't)	None	None	None	N/A
QR-01	Constrains UC-02	Listing, Textbook, Course	Listing Catalog	QAS-01	6.1, scenario QAS-01
QR-02	Constrains UC-01	Listing, Textbook, Course	Listing Creator	QAS-02	6.1, scenario QAS-02
QR-03	Constrains UC-03, UC-06	Payment	Order Manager, Payment Processor	QAS-03, CON-02	Spike S-01
QR-04	Constrains all UC	Listing (status)	All UI views	ADR-01	6.2 model view controller separation
QR-05	Constrains UC-03 (A2)	Listing, Reservation, ReservationDesk	OrderManager	ADR-01	UC-03 alternate flow A2
QR-06	Constrains all UC	All classes	Deployment host (6.4)	None	6.4 development view
CON-01	Precondition (UC-01, 03, 04, 05, 07)	Student	External: NetID SSO	CON-01	6.4 trust boundary
CON-02	Constrains UC-03, UC-06	Payment	Order Manager, External Payment Processor	QAS-03	Spike S-01
CON-03	None (schedule constraint)	None	None	None	Section 10.2 Plan to P3

9.2 Gap analysis

Gap	Items	What you did about it

Requirements with no component	FR-11 (shipping), CON-03 (P3 deadline)	FR-11 is "won't", CON-03 is a schedule constraint > no change
Components with no requirement, QAS or ADR	None	Every component in 6.3 traces to at least one QAS/UC
Use cases not analyzed (and why that is acceptable)	UC-02, UC-05, UC-06, UC-07	UC-02 and 05 are simple read operations, 06 is delegated to Stripe (S-01), UC-07 mirrors 04 for buyer and is still mapped to Handoff Tracker in 6.3 and 9.1

10 Project Plan (updated; was P1 section 4)

10.1 Roles and working agreement

Team member	Role for P1 to P2	How the team makes a decision when it disagrees
Rajit Ravindran	Architecture and Model designer	We will reach an agreement together as a group and if we can't agree on something, we will take a vote where the majority wins.
Abhishek Vemuri	Tester	We will discuss different opinions as a group and try to reach an agreement. If we cannot agree, we take a majority vote; if the vote is tied, the project manager decides.
Abran Garcia	Risk Manager	We will discuss it as a group to reach an agreement, and we will vote if we cannot come to an agreement.
Derrick Greene	Project manager	We will discuss different opinions as a group and try to reach an agreement. If we cannot agree, we will take a majority vote.
Steven Law	Analyst + Designer	We discuss it as a group first; if we still disagree, we take a majority vote, and the project manager breaks a tie.

Noah Patel	Reviewer	We will first discuss it as a group, and if an agreement cannot be made then we will take a vote, and the project manager will break ties.
Logan Tavares	Repository and integration organizer/reviewer	We will try to decide as a group and agree on a decision. If an agreement cannot be made, it will go to a majority vote.

10.2 Plan to P3

Milestone	Date	Owner	Done when
Finalize design and assign P3 tasks	Nov. 15, 2026	Derrick Greene / Rajit Ravindran	Architecture, diagrams, and team responsibilities are finalized.
Build and test BookBridge features	Nov. 30, 2026	Development team / Abhishek Vemuri	Listing, searching, reservations, and payment prototype are implemented and tested.
Final testing and P3 submission	Dec. 9,2026	Derrick Greene / Entire team	Final testing is completed, documentation is updated, and the project is submitted.

10.3 Risks (updated)

Risk	Status since P1	Likelihood / impact	Mitigation and owner
R1 — UTD SSO approval	Still open. Approval has not been confirmed, so the risk remains.	Medium / High	Confirm UTD SSO access before integration. If unavailable, propose a prototype alternative and document the requirement change. Owner: Rajit (proposed). ADR reference: pending.
R2 — Payment processor integration	Downgraded. A $25 Stripe	Medium / High	Use Stripe-hosted checkout to avoid handling

	sandbox payment succeeded, showing hosted checkout is feasible. Full integration remains untested.		card information directly. Test payment callbacks, failures, and reservation updates in P3. Owner: Derrick Greene. Evidence: S-01 Stripe spike, `docs/payment-feasibility.md`.
R6 — Payment and reservation consistency	New architectural risk. Payment could succeed while a reservation fails to update, or two buyers could attempt to reserve the same book.	Medium / High	Make sure only one student can reserve a book at a time and confirm that the Stripe payment went through before completing the purchase. Test what happens when multiple students try to reserve the same book or when a payment fails. **Owner:** Development team (to be confirmed). **ADR:** Pending.

10.4 Repository

Repository URL: https://github.com/rajitplay/CS-3354-Team-6

Folder Structure:

- `docs/p2/` — P2 report and supporting documents.
- `docs/p2/diagrams/` — P2 diagrams and models.
- `spikes/` — Architecture spike testing and results.
- `docs/payment-feasibility.md` — Stripe payment feasibility test.

Branch and Pull Request Rules: Team members should work on their own branches and submit pull requests for review before merging changes into the main branch.

Repository Access: The repository must be public or accessible to the instructor before the deadline.

P2 Tag: The `p2` tag must point to the commit containing the final P2 PDF.

10.5 Use of AI tools in P2

Member	Tool	What it was used for	What you checked or changed
Derrick Greene	ChatGPT	Helped organize and word my Stripe payment test report, architecture spike results, and project planning sections.	Verified the Stripe test results using the sandbox dashboard, checked the successful $25 payment, and reviewed the wording to make sure it matched the work I completed.
Steven Law	Claude (Anthropic)	Drafting: wrote first drafts of Sections 3.4, 4.1 to 4.3, 5.1 and 5.2 and drew the diagrams	Read each draft before adding it, checked it against the course lectures and our P1 document, and changed it where it did not match our P1 style and the template format

11 Open Questions (updated; was P1 section 5)

ID	Question	Raised by	Owner and due date	Status or answer	Assumption meanwhile
Q1	Will UTD allow us to use its sign-in system for BookBridge?	Derrick Greene	Rajit — Nov. 15, 2026 (proposed)	Open. UTD approval has not been confirmed.	The team will explore an alternative sign-in method for the prototype if UTD approval isn't available.
Q2	How will BookBridge confirm payments and update book reservations?	Derrick Greene	Development team — Nov.	Partially answered. Stripe sandbox	The team plans to use

			15, 2026 (proposed)	payment worked, but integration hasn't been tested.	Stripe payment confirmation before updating the reservation status.

12 Baseline v1.0 Declaration (was P1 section 6)

We declare this document, version 1.0, the baseline for P2. Sections 1 to 9 describe what we will build in P3. Any later change to them is recorded in section 0.2 with its reason.

Team member	Signature or initials	Date
Rajit Ravindran	RR	11/8/2026
Derrick Greene	DG	11/8/2026
Steven Law	SL	11/8/2026
Logan Tavares	LT	11/8/2026
Abran Garcia	AG	11/8/2026
Abhishek Vemuri	AB	11/8/2026
Noah,Patel	NP	11/8/2026